

NYCHA DEVELOPMENT DATA

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OUR CLEANED DATA

PROGRAM	TYPE	NUMBER OF RENTAL ROOMS	TOTAL POPULATION	DEVELOPMENT COST	PER RENTAL ROOM	AVG MONTHLY GROSS RENT	BOROUGH	COMMUNITY DISTRICT
1	1	868.5	413	\$5,031,405	\$5,341	\$488	1	6
1	2	293.5	141	\$2,205,187	\$7,302	\$502	1	3
1	1	350	157	\$1,880,013	\$4,694	\$514	2	10
1	1	386	116	\$2,278,928	\$5,814	\$517	1	9
1	1	157.5	65	\$4,503,296	\$28,412	\$698	2	7
1	1	1,101.00	480	\$4,703,702	\$4,272	\$543	3	3
1	1	530	213	\$4,290,021	\$7,981	\$805	2	3
1	1	725.5	330	\$3,631,814	\$4,992	\$631	2	7
1	1	4,305.00	2,043	\$17,878,818	\$4,149	\$547	1	1
1	1	3,751.50	1,708	\$8,951,000	\$2,356	\$618	3	8
1	1	1,824.50	848	\$4,666,000	\$2,540	\$605	3	8
1	1	5,120.00	2,111	\$12,262,000	\$2,391	\$619	2	7
2	1	762.5	336	\$7,053,000	\$9,250	\$607	2	7
1	1	5,564.50	2,646	\$14,122,000	\$2,519	\$715	4	1
1	1	1,155.00	493	\$10,797,972	\$8,489	\$651	3	2

PROGRAM:

1 = FEDERAL
2 = MIXED
FINANCE/LLC1
3= MHOP

BOROUGH:

1 = BRONX
2 = MANHATTAN
3 = BROOKLYN
4 = QUEENS
5 = STATEN ISLAND

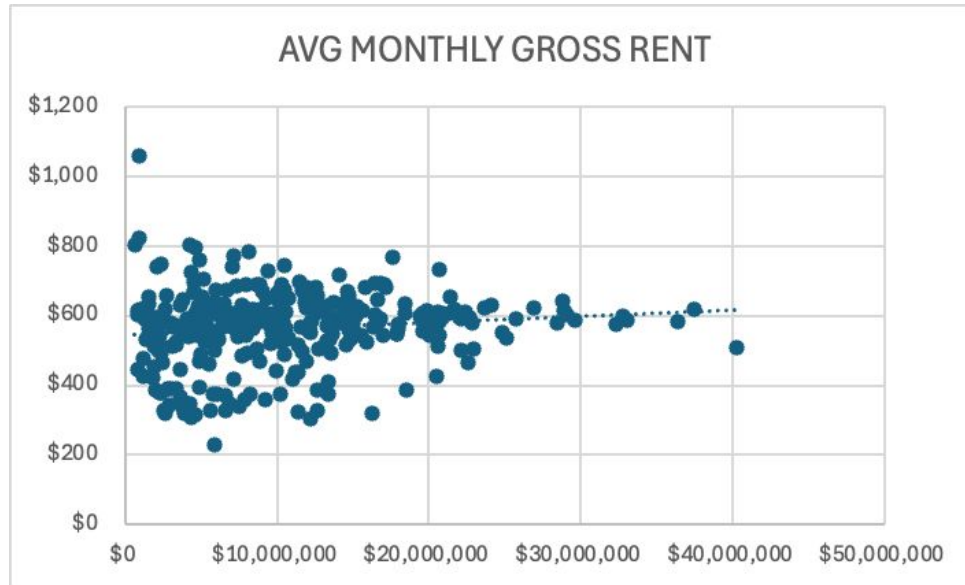
TYPE:

1 = NEW CONST
2 = REHAB

Using the NYCHA Development Data Book, can we determine when and where it's profitable to build affordable housing? (gross revenue only, excluding other data like capital and construction costs)

Independent variable: Development Costs

Dependent variable: Average Monthly Gross Rent (avg. monthly gross rent/development cost per rental)

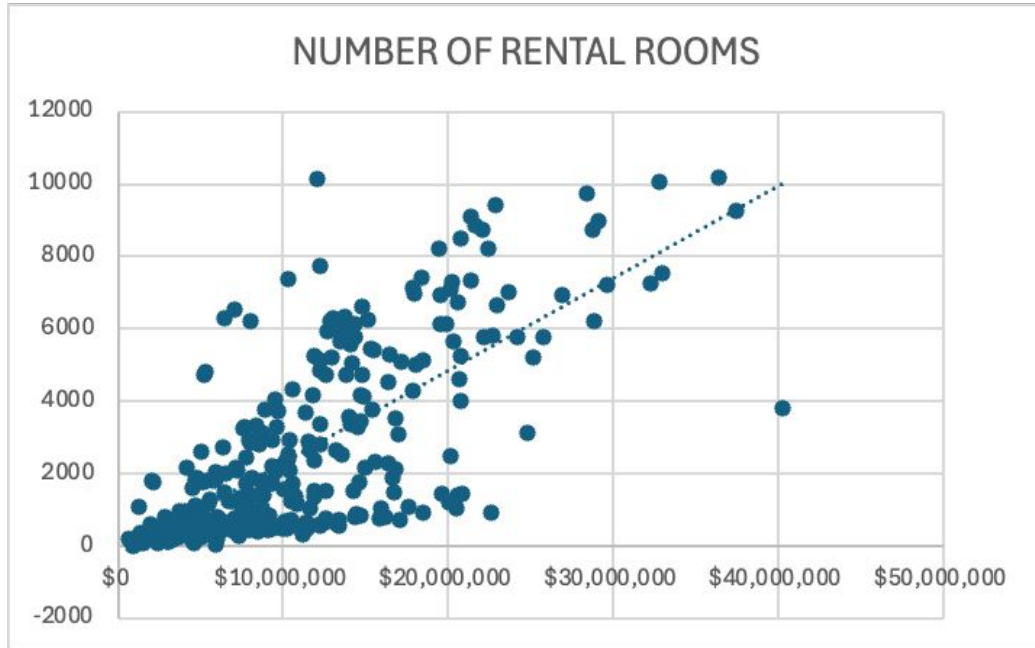


Low Correlation between development costs and avg. rent (-0.13), but F is under 0.05 and the P-value is statistically significant, it should not be ignored. There is an outlier in the lower costs as well.

Development costs and Number of Rental Rooms

Independent variable: Development costs

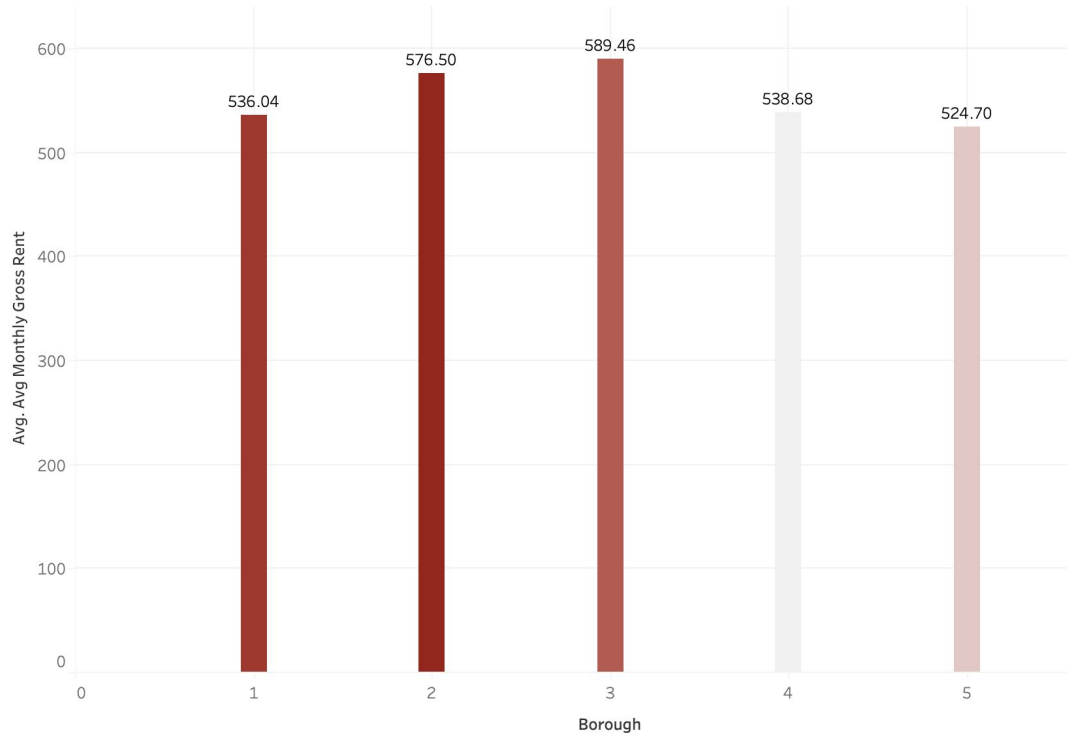
Dependent variable: Number of rental rooms



Strong correlation between development costs and number of rooms.(0.73)

Which borough looks the most attractive for subsidized developers?

Average Gross Rent (by borough) vs. Development Cost per Rental



The Bronx and Manhattan have the highest development costs per rental room without commanding premium rental prices.

Brooklyn and Queens have marginally better gross rental prices, but Queens boasts half the development cost per room than Manhattan.

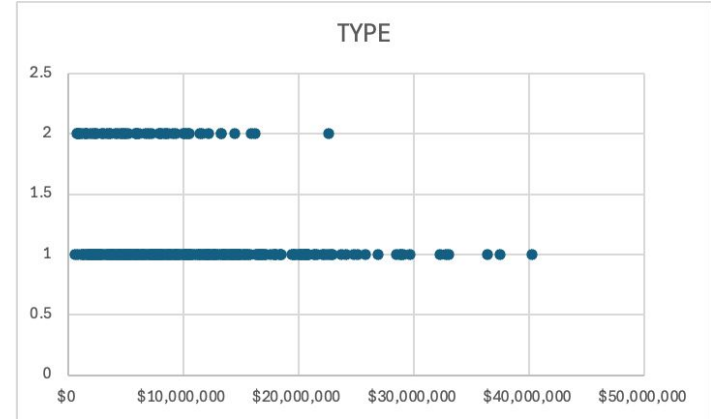
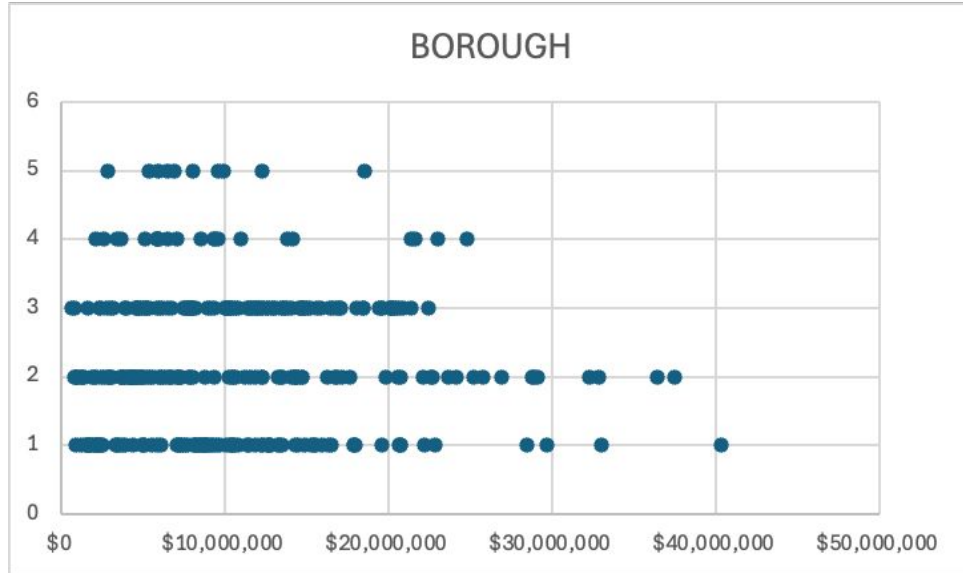
Which borough looks the most attractive for subsidized developers?

Independent variable: Borough

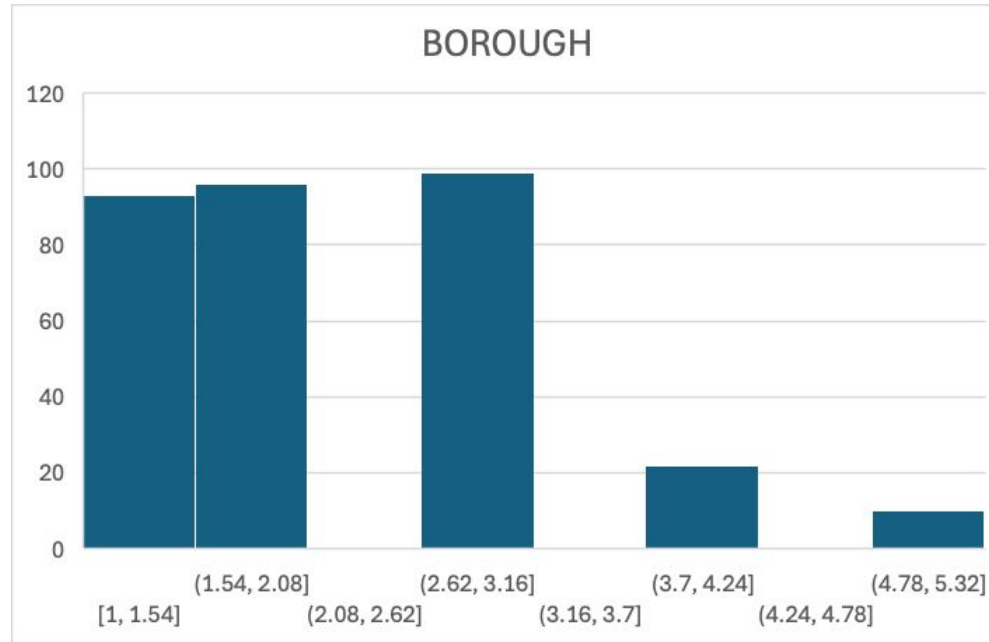
Dependent variable: Development Costs

Weak correlation between Borough and development costs. (0.0014)

New construction vs. rehabilitation (building type) correlation was weak as well but F and the P were values under 0.05 meaning it is statistically significant. Rehabbed building on average are cheaper.



Which boroughs have the most and least developments?



Bronx - 93, Manhattan - 96, Brooklyn - 99, Queens - 22, Staten Island - 10

Which is more profitable between new construction vs rehabilitation?

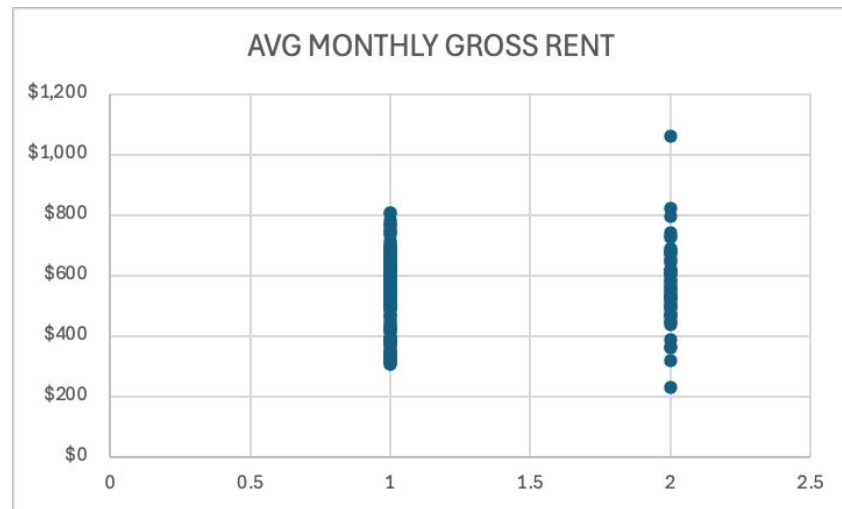
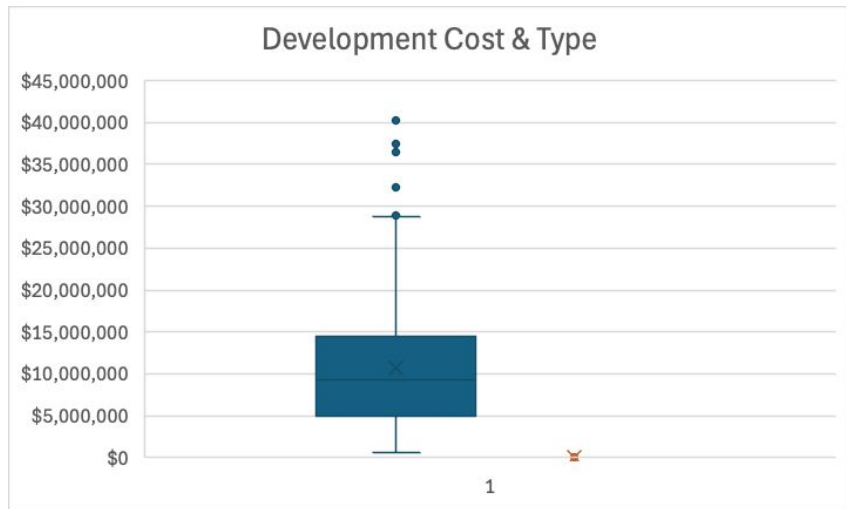
Independent variable: Type

Dependent variable: Profitability (development cost, average monthly gross rent)

Weak correlation between development costs and Type.
(-0.24)

Average development cost is under \$9.3m but new construction can cost more than rehab buildings.

Outside of a few outliers from rehab buildings, they make about the same as new construction in gross revenue but are cheaper to develop.

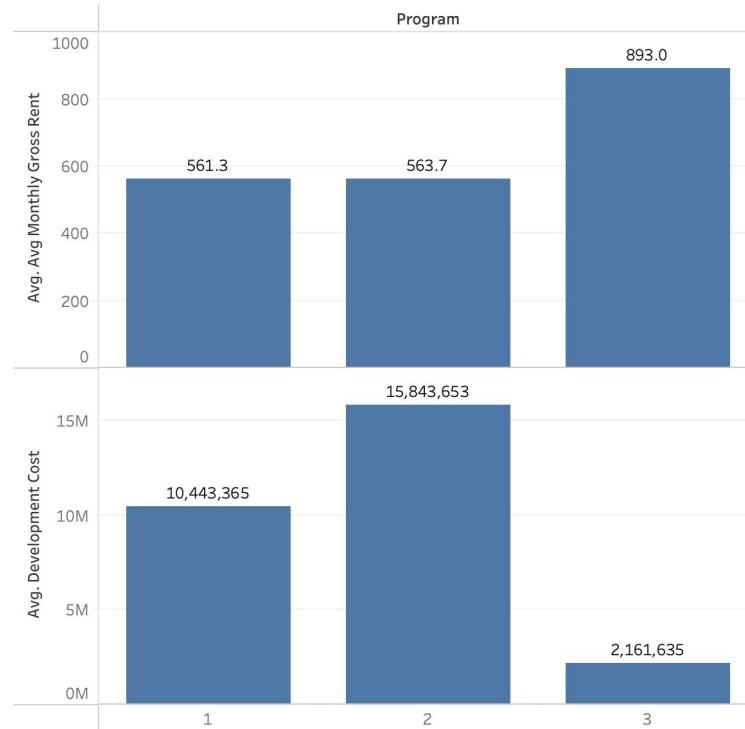


Which financing program is the most successful?

Independent variable: Program

Dependent variable: Profitability

Average Development Cost (by program) vs. Average Monthly Rent



Weak correlation between development cost and program (0.06), but MHOP has the lowest overall average development costs.

MHOP (3) are the most profitable type of buildings by lowest development costs and highest average gross rent.

OUR RECOMMENDATIONS

There is no significant difference in gross profits across the boroughs or communities, but there are areas for development. The Bronx and Manhattan have the most costly rentals.

Most subsidized housing is being developed in Brooklyn, Manhattan, and the Bronx, while Queens and Staten Island have the least number of developments.

Queens is actually the most attractive borough because it has no rehabilitation projects. Rehab projects cost less, and the average gross rent is on par with new construction.

Multi-Family Home Ownership Programs (MHOP) are the most cost-effective and have the highest average gross rent, but there are fewer of these programs compared to other options. Our suggestion would be to focus on new construction under the Multi-Family Home Ownership Program, or to rehabilitate existing apartment buildings in Queens and Staten Island where there is less construction.